

Propiedades de la dilatación isotrópica

Observación 1. Si $\phi \in \mathcal{L}^1(\mathbb{R}^n)$, entonces $\lim_{M \rightarrow \infty} \int_{\{\|x\| > M\}} \phi(x) \, dx = 0$ ya que

$$\lim_{M \rightarrow \infty} \int_{\{\|x\| > M\}} \phi(x) \, dx = \lim_{M \rightarrow \infty} \int_{\mathbb{R}^n} \phi(x) \mathbb{1}_{\{\|x\| > M\}}(x) \, dx \stackrel{\text{TCD}}{=} \int_{\mathbb{R}^n} \phi(x) \cdot 0 \, dx = 0.$$

Ahora sea $t > 0$, definimos $\phi_t(x) := t^{-n} \phi\left(\frac{x}{t}\right)$. Entonces,

$$(1) \quad \int_{\mathbb{R}^n} \phi_t(x) \, dx = \int_{\mathbb{R}^n} \frac{1}{t^n} \phi\left(\frac{x}{t}\right) \, dx = \int_{\mathbb{R}^n} \phi(y) \, dy \text{ mediante el cambio } y = \frac{x}{t}.$$

$$(2) \quad \forall \eta > 0 : \lim_{t \rightarrow 0^+} \int_{\{\|x\| \geq \eta\}} \phi_t(x) \, dx = \lim_{t \rightarrow 0^+} \int_{\{\|y\| \geq \eta/t\}} \phi(y) \, dy = 0.$$